

OTIS ANGB, MA, USA

WMO: 725061

Lat: 41.650N

Lon: 70.517W

Elev: 130

StdP: 14.63

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 14704

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.5	14.0	-7.9	3.6	12.5	-2.3	4.9	18.2	32.6	40.0	30.4	35.0	11.0	300	0.577

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.9	84.0	73.5	81.6	71.6	79.2	69.6	75.6	81.3	74.3	78.8	72.9	76.7	11.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.4	124.7	77.4	72.8	122.3	76.7	72.1	119.4	76.0	39.3	80.5	37.7	78.9	36.8	76.8	80.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.9	24.2	20.8	DB	2.2	88.7	7.5	3.1	-3.3	90.9	-7.7	92.8	-11.9	94.5	-17.4	96.8	(4)
(5)			WB	0.6	77.7	6.9	1.9	-4.4	79.1	-8.5	80.3	-12.4	81.3	-17.4	82.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.7	31.1	31.7	36.7	46.2	56.2	63.5	71.2	70.1	63.9	54.7	44.6	37.0
	DBStd	15.73	9.13	8.53	8.00	6.67	6.32	5.43	4.56	4.64	6.14	7.28	7.79	8.89
	HDD50	2329	588	514	418	150	14	0	0	0	0	36	197	412
	HDD65	5719	1052	933	878	564	283	91	5	9	94	327	613	870
	CDD50	2568	1	1	5	36	205	404	657	622	416	181	33	7
	CDD65	483	0	0	0	0	9	45	197	165	60	7	0	0
	CDH74	2311	0	0	0	6	57	210	985	821	217	15	0	0
	CDH80	415	0	0	0	0	9	35	205	142	23	1	0	0
Wind	WSAvg	9.9	10.8	11.2	11.2	10.8	9.8	9.1	8.7	8.0	8.6	10.0	10.2	10.1
Precipitation	PrecAvg	46.8	3.9	3.3	4.8	4.3	3.8	3.3	2.8	3.7	4.3	3.7	4.2	4.8
	PrecMax	59.5	6.3	7.3	10.3	7.9	6.7	11.0	7.3	7.6	12.8	10.0	6.9	9.4
	PrecMin	38.8	1.9	1.8	0.6	1.5	1.2	0.4	0.7	0.3	1.3	0.8	0.7	1.5
	PrecStd	5.0	1.3	1.4	2.2	1.6	1.4	2.4	1.4	1.7	2.7	2.3	1.8	2.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.5	55.3	63.2	72.3	80.9	84.6	88.4	86.4	82.3	75.3	65.9	62.6
		MCWB	55.3	50.9	55.1	59.9	68.2	71.9	76.3	75.4	72.8	66.3	62.3	60.3
	2%	DB	52.1	51.8	55.2	64.3	73.4	79.4	84.2	83.7	79.2	71.9	63.0	57.1
		MCWB	51.0	49.6	49.5	56.4	64.1	67.0	73.9	73.0	70.0	66.3	59.6	55.2
	5%	DB	48.1	47.9	51.5	60.7	70.1	76.5	81.9	81.2	76.6	68.1	59.4	54.4
		MCWB	46.5	45.4	47.1	52.6	61.3	65.9	72.6	71.4	69.6	64.7	55.9	52.4
	10%	DB	45.4	44.8	47.8	56.7	66.1	73.0	79.4	78.8	73.2	65.8	57.0	50.4
		MCWB	42.8	41.7	44.4	49.6	58.9	64.9	70.3	70.0	67.9	63.2	53.6	48.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.2	54.3	56.7	63.6	70.0	73.6	77.9	77.6	75.3	70.2	63.8	60.5
		MCDB	56.1	55.2	59.9	67.1	78.2	83.4	85.7	83.6	80.5	72.1	64.5	61.4
	2%	WB	50.8	49.6	51.5	59.2	66.3	70.9	75.8	75.8	73.2	67.4	60.7	55.8
		MCDB	51.5	50.3	54.6	62.9	71.9	76.6	82.3	80.8	76.4	69.0	62.0	56.8
	5%	WB	46.5	45.8	47.8	54.9	63.5	69.1	74.4	74.3	71.9	65.5	57.5	52.5
		MCDB	47.2	47.1	50.0	58.7	66.4	73.1	79.3	78.4	75.0	67.5	59.8	53.8
	10%	WB	43.1	41.4	44.8	51.7	61.1	67.2	73.1	72.8	70.0	63.2	54.5	48.7
		MCDB	45.1	43.9	46.7	54.6	65.3	70.9	76.9	76.4	72.7	64.8	56.7	50.0
Mean Daily Temperature Range	5% DB	MDBR	14.1	14.4	14.3	15.8	15.2	14.4	13.9	14.5	15.2	15.2	14.7	13.8
		MCDBR	18.1	18.1	18.2	21.0	20.5	18.9	16.1	16.5	16.7	16.6	16.7	16.9
		MCWBR	18.2	16.3	13.7	13.2	11.5	10.1	7.8	8.4	10.1	11.7	14.4	17.2
	5% WB	MCDBR	17.3	18.0	17.2	18.6	17.5	15.3	14.1	13.7	14.0	14.1	14.6	17.0
		MCWBR	18.0	17.6	14.7	14.1	11.8	9.8	7.8	7.8	9.5	11.8	14.7	17.6
Clear-Sky Solar Irradiance	taub	0.307	0.313	0.336	0.373	0.409	0.460	0.483	0.437	0.373	0.345	0.317	0.305	
	taud	2.513	2.500	2.474	2.394	2.337	2.230	2.173	2.292	2.451	2.533	2.568	2.563	
	Ebn at Noon	269	285	290	285	276	261	253	262	274	270	264	261	
	Edh at Noon	23	27	31	36	39	44	46	40	32	26	22	20	
All-Sky Solar Radiation	RadAvg	563	847	1168	1507	1695	1821	1884	1693	1375	937	633	477	
	RadStd	44	83	89	161	164	168	104	88	96	80	43	49	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (58 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+169	+81

Nomenclature: See separate page